LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART II

SESSION 10

Financial Stability & Corporate Finance Conditions



From markets, to macro, to decisions that matter — in Nigeria and the world.

❐ Reading: Risk Management at Silicon Valley Bank (F-0997-E) · Facilitator: Elvis Ozegbe

RECAP

Session 9 in Three Ideas

Debt dynamics

Debt ratios rise when the interest rate beats growth (r > g) and the primary balance is in deficit.

Revenue problem

Nigeria's debt is moderate relative to GDP but heavy relative to revenue: interest eats a large share of what government collects.

Fiscal–monetary link

When government borrows from the central bank, fiscal problems become inflation problems.

TODAY'S PLAN

Session 10 at a Glance — 75 Minutes

Why 'safe' government bonds can sink a bank — and what that means for the cost of your firm's credit.



Recap & Hook

How did a bank holding safe bonds fail?

Yields & Bond Prices

The yield curve, inversion and the price–yield seesaw

Bank Fragility

Duration gap, the SVB story, macroprudential policy

Lab: Stress-Test a Bank

Could our bank be the next SVB?

BY THE END OF TODAY

Session 10 Learning Objectives

You will be able to:

01

Read the Yield Curve

Explain what the level and slope of the FGN yield curve say about inflation, policy and the cost of long-term money.

02

Spot Interest-Rate Risk

Use duration to estimate how much a bond or a bank loses when rates rise.

03

Link Stability to Your Firm

Explain how bank fragility and macroprudential rules change credit availability, spreads and investment.

THE HOOK

The Bank That Bought Safe Bonds

“We hold government bonds. They never default. What could go wrong?”

Silicon Valley Bank (SVB) failed in March 2023 without a single government bond defaulting. Three things combined:

Rates rose fast

US policy rates went from near zero to about 5% in a year. Bonds bought at low yields lost value.

Long assets, short funding

SVB had put deposits — which can leave overnight — into long-dated bonds.

Depositors ran

Once losses became public, uninsured depositors withdrew billions in a day. Selling bonds turned paper losses into real ones.

NIGERIAN BUSINESS CASE

Could a Nigerian Bank Fall Into the SVB Trap?

The situation

A mid-sized Nigerian bank holds **₦1,850bn of FGN bonds** bought around 13%, funded mostly by current and savings deposits. Yields jumped in 2024 as the CBN tightened.

The decision question

How much **economic value** has the bank lost, how big a rate shock could wipe out its equity, and what should the board do?

The data we have

bank_balance_sheet.csv — a fictional bank's assets and liabilities with durations and rates. ng_yield_curves.csv — five stylised FGN yield curves (2021–2025). *Stylised teaching data.*

What a good answer looks like

A **stress test** in naira: equity left after +1 to +5 point shocks, and one or two concrete actions (shorten the bond book, hedge, raise capital, lengthen funding).

PART

01

Yields and Bond Prices

The seesaw that every treasurer must understand.

What a Yield Curve Shows

IN PLAIN ENGLISH

The yield curve plots the interest rate the government pays to borrow for **3 months, 1 year, 10 years** and longer. Its **level** tells you how expensive money is; its **slope** tells you what markets expect next.

EXAMPLE

In 2021 the curve was **low and steep** (2.5% for 3 months, 12.8% for 10 years). By mid-2024 it was **high and inverted** (17.5% for 3 months, 21% for 1 year, 18.8% for 10 years).

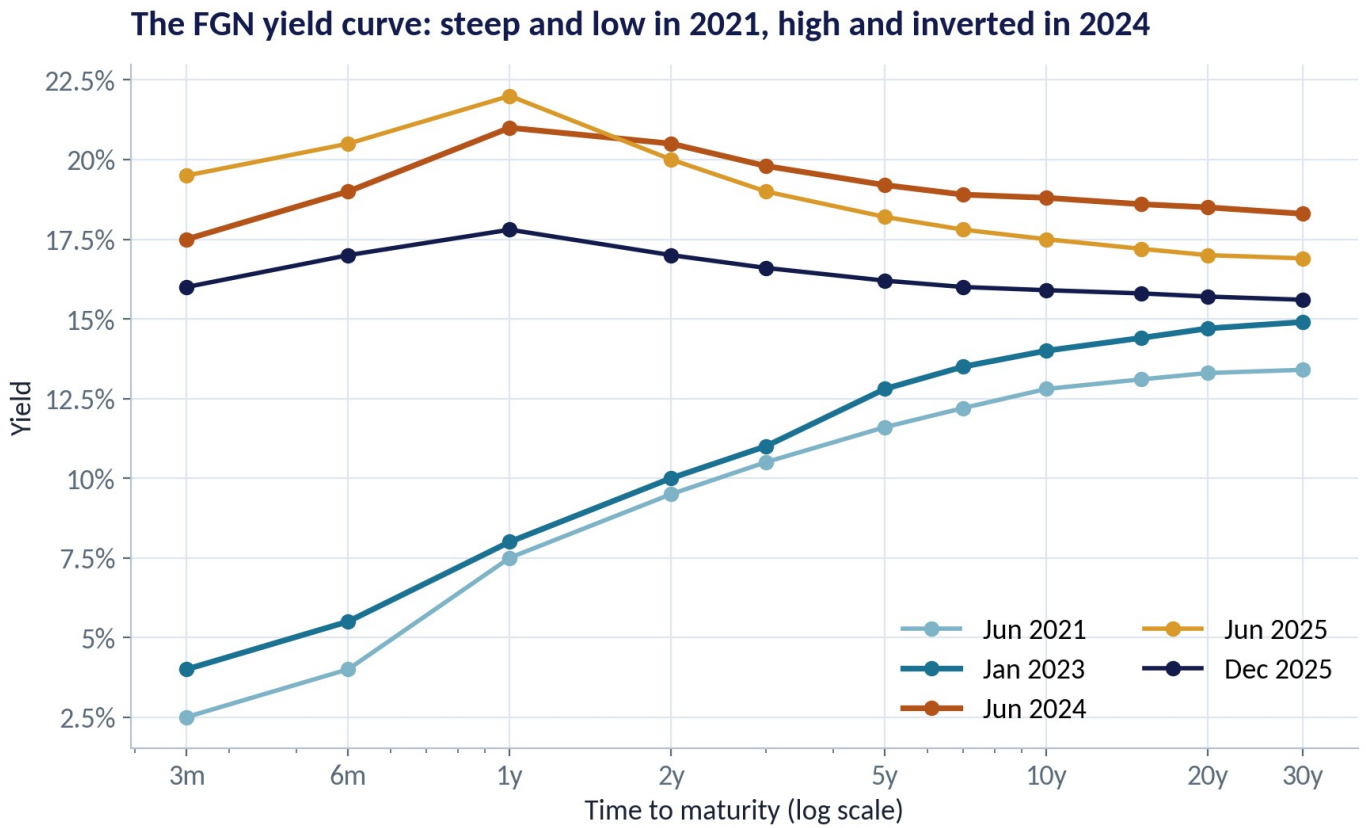
THE FORMULA

$$\text{long yield} \approx \text{average expected short rate} + \text{term premium}$$

WHY A MANAGER CARES

An inverted curve says markets expect rates to fall later — often because tight policy will slow the economy. It also tells you when to borrow long and when to wait.

From Steep and Low to High and Inverted



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F01 · replicate: s10_figures.R / .py

WHAT TO SEE

- **2021**: cheap short-term money, long rates well above short — a normal, steep curve.
- **Mid-2024**: the CBN's tightening pushed short rates above long rates — an **inverted** curve.
- **End-2025**: lower and flatter as inflation eased and rate-cut expectations built.
- For a CFO the level is the cost of funds; the slope hints at whether to lock in long or stay short.

When Yields Rise, Bond Prices Fall

IN PLAIN ENGLISH

A bond pays fixed coupons. If new bonds pay more, nobody will pay full price for your old, lower-coupon bond — so its **price falls** until its yield matches the market.

EXAMPLE

A 10-year FGN bond with a 13% coupon is worth **100** at a 13% yield, **89.8** at 15%, **77.2** at 18% and **70.2** at 20%.

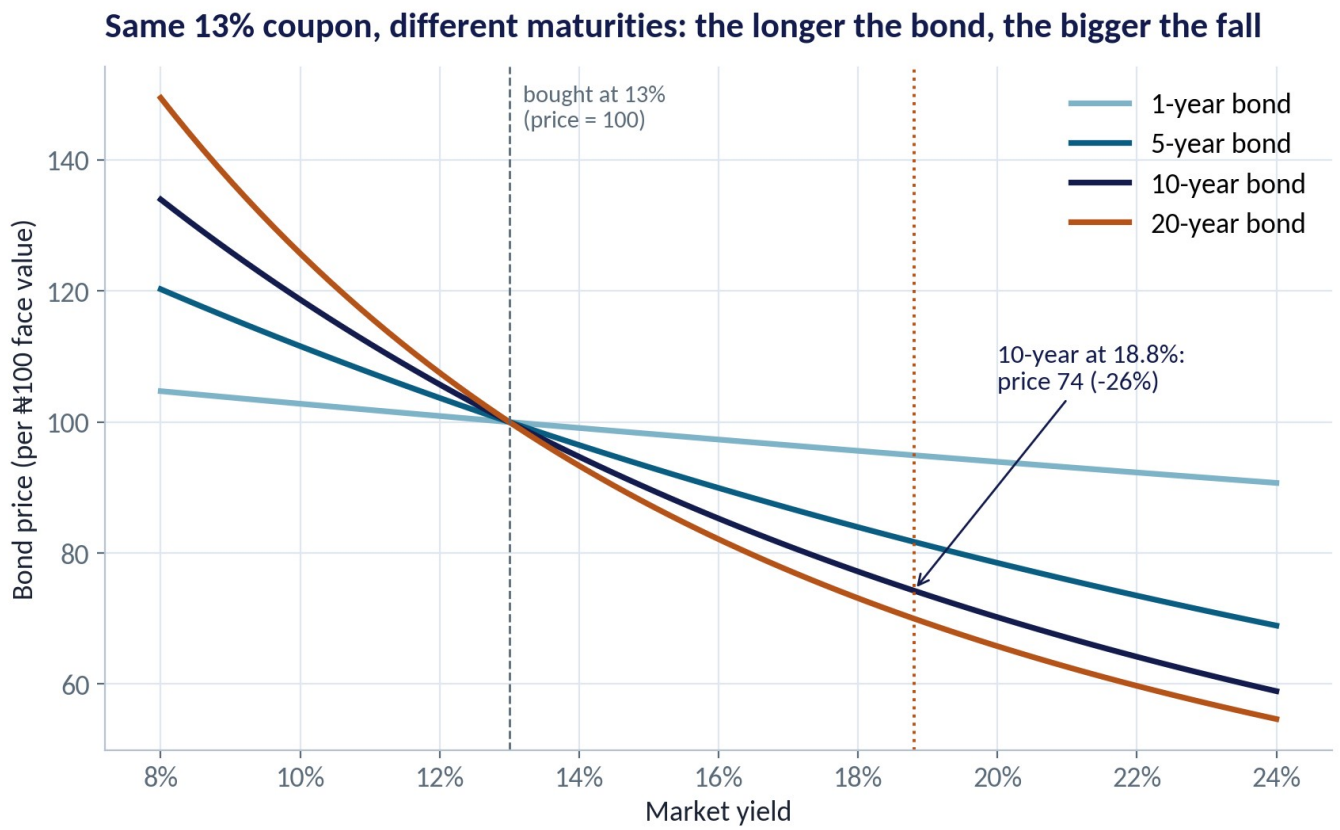
THE FORMULA

$$\% \text{ price change} \approx - \text{modified duration} \times \text{change in yield}$$

WHY A MANAGER CARES

Duration — roughly the average time until you get your money back — measures sensitivity. The longer the bond, the bigger the fall when rates rise.

Long Bonds Fall Most



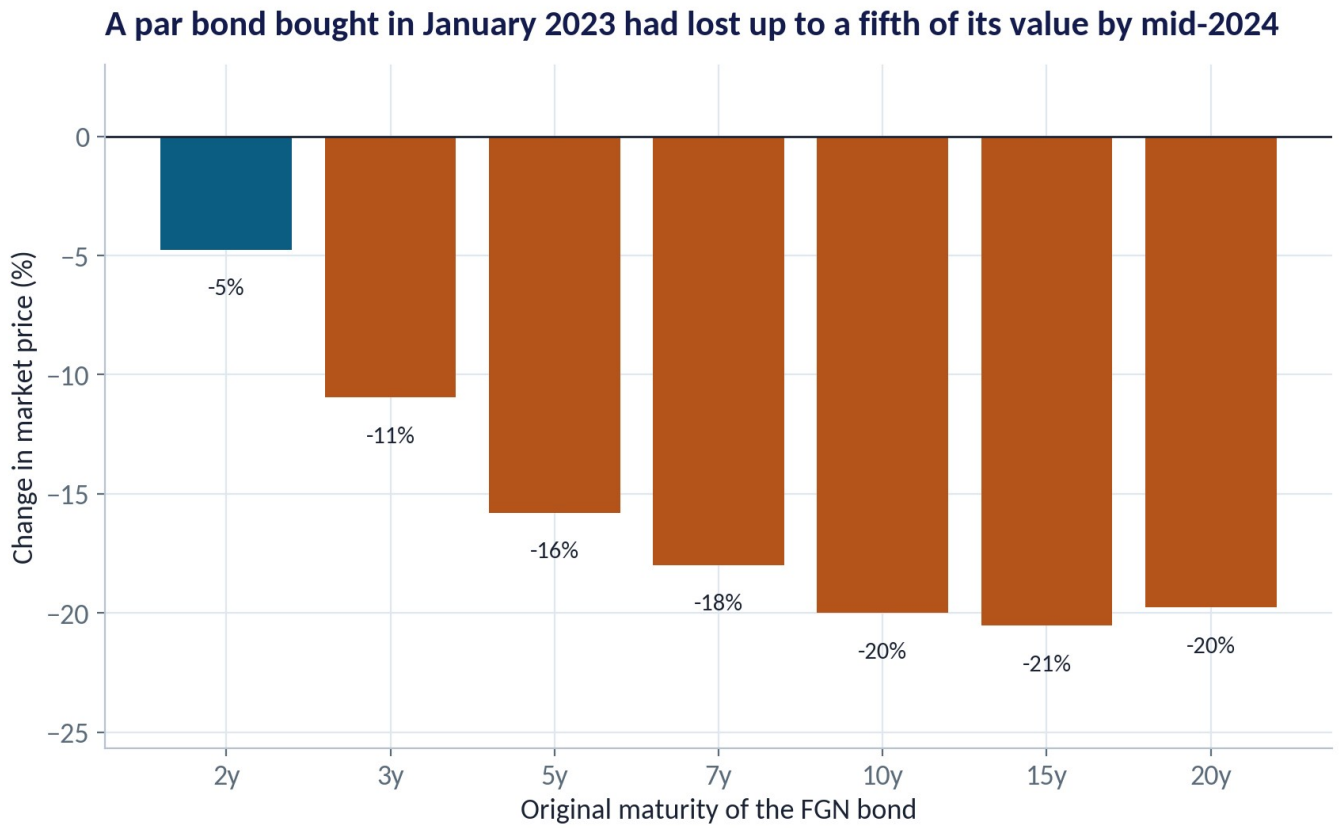
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F02 · replicate: s10_figures.R / .py

WHAT TO SEE

- All four bonds pay a 13% coupon and are priced at 100 when the yield is 13%.
- At an 18.8% yield the **1-year bond barely moves**; the **10-year bond falls to about 74.**
- The **20-year bond** falls further still.
- Rule of thumb: more duration = more interest-rate risk.

What Happened to Bonds Bought in January 2023



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F03 · replicate: s10_figures.R / .py

WHAT TO SEE

- Take bonds bought at par in January 2023 and value them in June 2024.
- Short bonds lost about **5%**; 10- and 15-year bonds lost about **20%**.
- No default, no missed coupon — the loss comes purely from **higher market yields**.
- If you must sell before maturity, that loss becomes real.

Quick Check: What Would You Say?

?

Your pension fund holds a 20-year FGN bond. The CBN signals more rate hikes. What is the most likely effect on the bond's market value?

A. It rises, because higher rates mean higher coupons on the bond you already hold

B. It falls, and by more than a 2-year bond would

C. It is unchanged until the bond matures

D. It falls by the same percentage as a 2-year bond

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

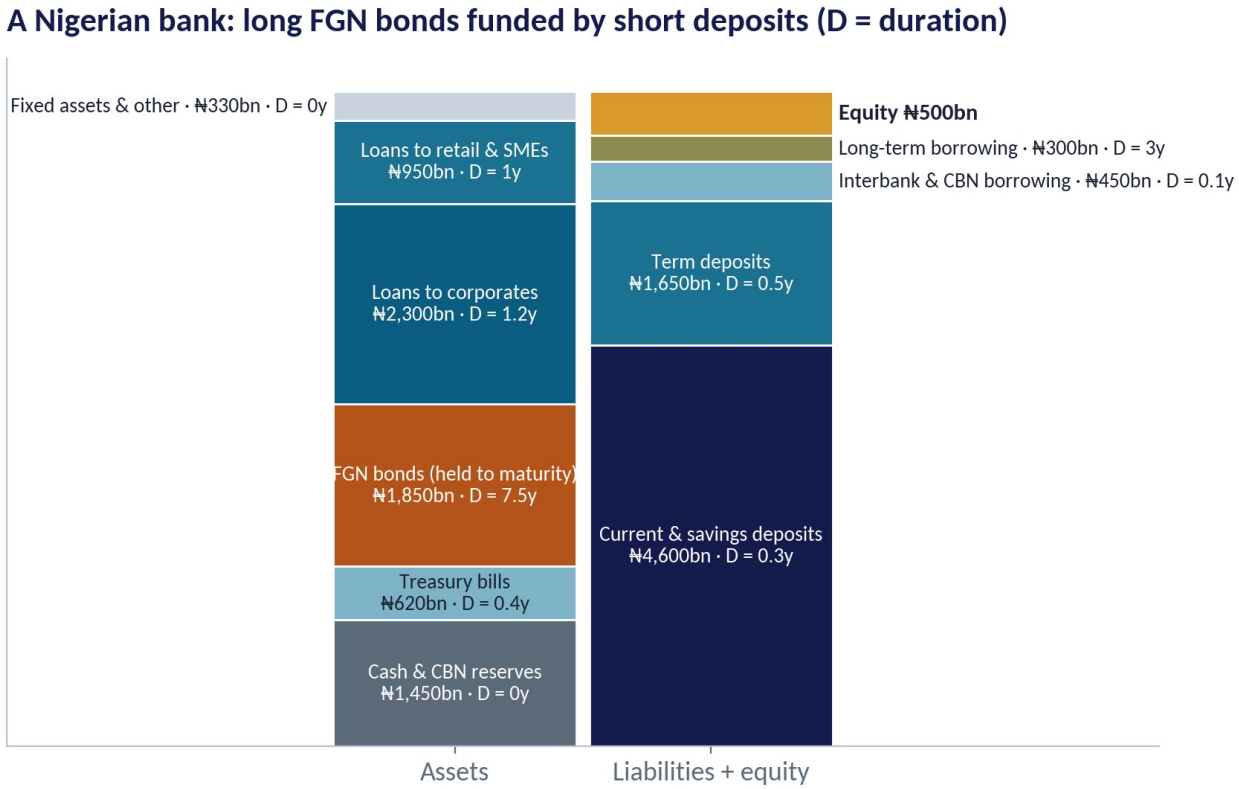
PART

02

Why Banks Are Fragile

Borrow short, lend long — and hope rates behave.

Long Bonds Funded by Short Deposits



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F04 · replicate: s10_figures.R / .py

WHAT TO SEE

- Assets **₦7,500bn**; liabilities **₦7,000bn**; equity only **₦500bn** — leverage of **15×**.
- The **FGN bond book (₦1,850bn)** has a duration of 7.5 years.
- Most funding is **current and savings deposits** with a duration of a few months.
- Banks earn money by this mismatch — and take risk because of it.

THE KEY NUMBER

The Duration Gap

IN PLAIN ENGLISH

The **duration gap** compares how long the bank's assets are with how long its liabilities are. A positive gap means asset values fall **more** than liability values when rates rise — so equity shrinks.

EXAMPLE

Asset duration **2.38 years**, liability duration **0.45 years** → gap **1.96 years**. A +1 point rise in rates cuts economic equity by about **₦124bn**.

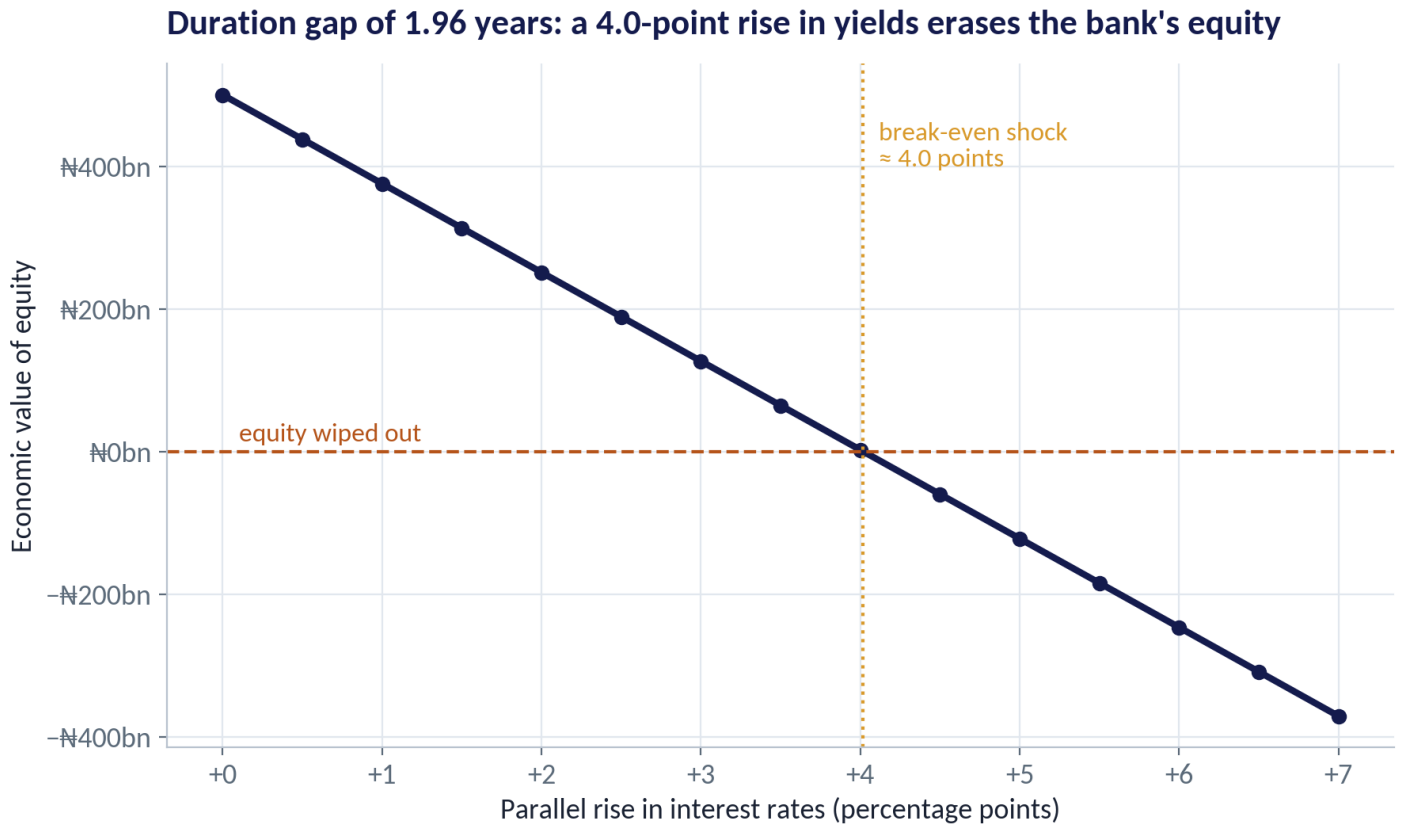
THE FORMULA

$$\text{gap} = D(\text{assets}) - (\text{liabilities} \div \text{assets}) \times D(\text{liabilities})$$

WHY A MANAGER CARES

The gap turns a rate forecast into a naira number the board understands: how much of our capital a rate shock would destroy.

How Big a Rate Shock Can the Bank Survive?



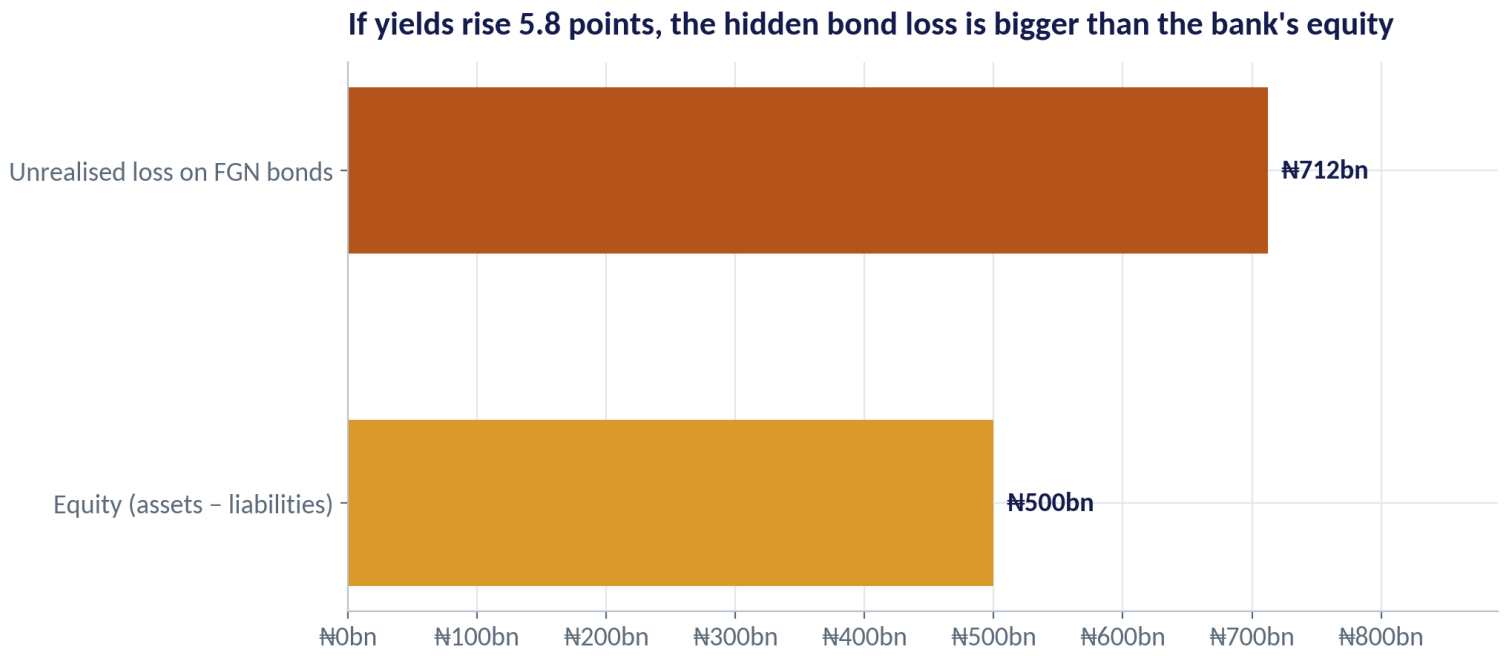
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F05 · replicate: s10_figures.R / .py

WHAT TO SEE

- Each point of higher rates removes roughly **₦124bn** of economic equity.
- +2 points leaves about **₦251bn**; +3 points about **₦127bn**.
- A rise of about **4 points** wipes out equity entirely.
- Nigerian 10-year yields rose by almost **5 points** between early 2023 and mid-2024.

The Hidden Loss Can Exceed Equity



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F06 · replicate: s10_figures.R / .py

WHAT TO SEE

- Using the 2024 yield curve, the bond book's unrealised loss is about **₦712bn**.
- That is **larger than the bank's ₦500bn equity**.
- Accounting rules for 'held-to-maturity' bonds can keep this loss off the income statement.
- It becomes real only if the bank must **sell** — for example, in a deposit run.

Micro-Prudential vs Macroprudential Policy

Micro-prudential

Is each bank safe on its own?

- Minimum capital and liquidity ratios
- Supervision of individual loan books
- Stress tests bank by bank
- Deposit insurance (NDIC in Nigeria)

Macroprudential

Is the system as a whole safe?

- Counter-cyclical capital buffers
- Limits on FX lending and open positions
- Cash reserve requirements and loan-to-deposit rules
- Recapitalisation drives when the economy changes

The CBN's 2024 bank recapitalisation programme is a macroprudential response: inflation and devaluation had shrunk the real value of banks' capital.

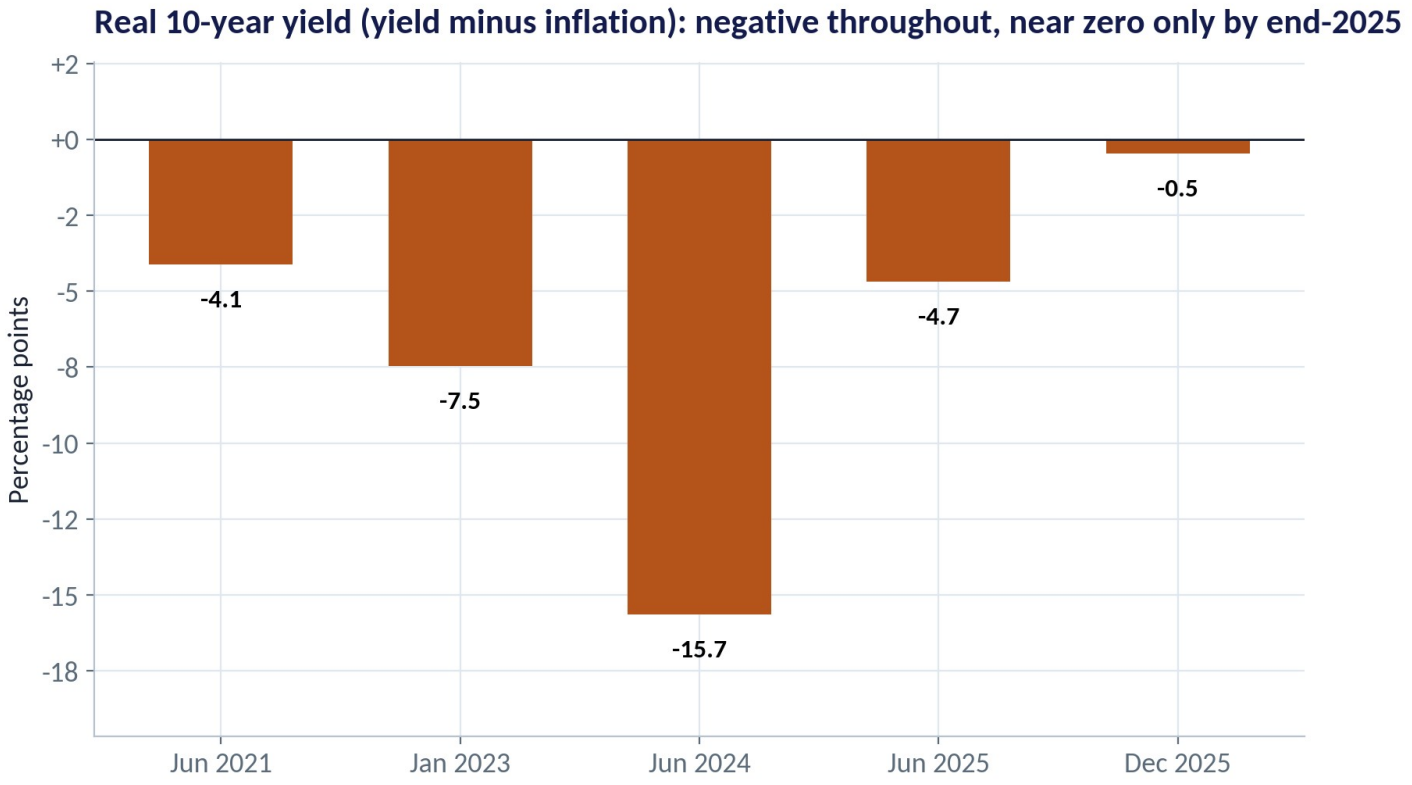
PART

03

From the Bank to Your Balance Sheet

Financial conditions decide who gets credit, and at what price.

Were Lenders Paid for Inflation?



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S10-F07 · replicate: s10_figures.R / .py

WHAT TO SEE

- Real 10-year yield = nominal yield − inflation.
- It was **deeply negative** in 2024 (about −16 points).
- By end-2025 it was close to zero (about **−0.5**).
- Negative real yields push savers to dollars and real assets; positive ones bring them back to naira bonds.

How Financial Conditions Reach Your Company

Price of credit

Lending rates follow the yield curve plus a **credit spread** for your risk. When yields hit 20%, prime borrowers paid well above that.

Quantity of credit

Banks short of capital or liquidity **ration** credit: smaller limits, shorter tenors, more collateral.

Investment hurdle

A higher cost of capital kills marginal projects. A plant that earned 22% looked fine at 14% funding, not at 25%.

Currency and tenor

Firms that switch to cheaper dollar loans swap interest-rate risk for **FX risk** — Session 11.

How Treasurers and Boards Get Caught Out

THE TRAP	THE BETTER HABIT
✗ “Government bonds are risk-free, so we can hold as many as we like.”	✓ They are free of **default** risk, not of **interest-rate** risk. Measure duration.
✗ Ignoring unrealised losses because they are 'held to maturity'.	✓ Report **economic value** to the board as well as the accounting value.
✗ Funding long assets with overnight money because it is cheap today.	✓ Match tenors, or **hedge** the gap.
✗ Assuming deposits are sticky.	✓ Mobile banking lets deposits leave in hours. Stress-test **liquidity** as well as capital.
✗ Borrowing short at the peak of an inverted curve without a plan.	✓ Decide deliberately between fixed and floating, short and long.

CONCEPT CHECK

Quick Check: What Would You Say?

?

The case bank's duration gap is 1.96 years and its equity is ₦500bn. A colleague says: 'A 2-point rise in yields is nothing for us.' What is the best reply?

A. Correct — the bonds will pay back at maturity, so there is no loss

B. A 2-point rise would cut economic equity by about ₦250bn, half our capital

C. A 2-point rise would wipe out all our equity

D. We cannot say without the bank's profit figures

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

Lab: Could Our Bank Be the Next SVB?

YOU WILL NEED

Data

`data/bank_balance_sheet.csv`
`data/ng_yield_curves.csv`

R script

`sessions/s10/R/s10_lab.R`

Python script

`sessions/s10/python/s10_lab.py`

20 minutes

1. Tabulate the five yield curves: which are **normal** and which **inverted**?
2. Price a 10-year, 13% bond at yields of 13%, 15%, 18% and 20%.
3. Compute the bank's **equity and leverage**.
4. Compute asset duration, liability duration and the **duration gap**.
5. Stress-test equity for +1, +2, +3 and +5 point rate shocks.

Deliverable: two sentences for the board on the bank's interest-rate risk and one action you would take.

WORDS TO OWN

Today's Vocabulary in Plain English

Yield	The annual return on a bond if bought at today's price and held to maturity.
Yield curve	Yields on government debt plotted against maturity.
Inverted curve	Short-term yields above long-term yields.
Term premium	Extra yield investors demand for lending long.
Duration	Weighted average time to a bond's cash flows; its rate sensitivity.
Duration gap	Asset duration minus leverage-adjusted liability duration.
Mark-to-market	Valuing an asset at today's market price.
Held to maturity	Accounting label that keeps bond losses off the income statement.
Credit spread	Extra interest a borrower pays above the government yield.
Macroprudential	Rules that protect the financial system as a whole.

KEY TAKEAWAYS

Session 10 in 5 Ideas

01 The **yield curve's level** is the price of money; its **slope** is the market's view of what comes next.

02 When yields rise, **bond prices fall** — long bonds most. Government bonds carry interest-rate risk.

03 Banks are fragile because they **borrow short and lend long** with thin equity.

04 The **duration gap** turns a rate shock into a naira loss of capital.

05 Financial conditions reach your firm through the **price and quantity of credit**.



NEXT › Session 11: Exchange Rates & Capital Flows

Before next session: Read the Thailand: An Imbalance of Payments case. Note the exchange-rate regime, the size of short-term foreign borrowing, and what happened to reserves in 1997.